

King's College London
 Department of Political Economy
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(Undergraduate & Taught Postgraduate)

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This assignment is entirely my own work. Quotations from secondary literature are indicated by the use of inverted commas around ALL such quotations AND by reference in the text or notes to the author concerned. ALL primary and secondary literature used in this piece of work is indicated in the bibliography placed at the end, and dependence upon ANY source used is indicated at the appropriate point in the text. I confirm that no sources have been used other than those stated.

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Candidate no.	A	G	0	4	2	0	7
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Module Title:	MSc Economics and Policy Dissertation (25~26 NSY)
Module Code (e.g. 4SSPP103):	7SSPN407
Assignment title:	Research Proposal
Deadline:	27th February 2026 3:00pm
Date Submitted:	27th February

Word Count: For details of what is included in the word count, and penalties incurred by exceeding the word count limit, please consult the handbook.	1152
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(i) I have used AI to generate ideas or structure suggestions, for assistance with understanding core concepts, or other substantial foundational and preparatory activity	☐
<i>If you have checked the above box please disclose AI tool(s), links and/or details of how AI was used [INSERT HERE]</i>	
(ii) I have used AI to generate some other aspect of the submitted assessment in accordance with the Department's/convener's specification of acceptable use	☐
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1 Introduction

This dissertation investigates the causal effects of generative artificial intelligence (GenAI) on firm-level outcomes. The analysis focuses on firm performance, labour demand, and particularly on share buybacks. The research uses an empirical, quasi-experimental design that exploits the public release of ChatGPT in November 2022 as a common technology shock, which is widely recognised as a turning point in the diffusion of AI. While AI technologies existed before 2022, ChatGPT made advanced GenAI tools widely accessible, inexpensive, and usable by a broad range of firms.

A central objective of the dissertation is to distinguish between active AI adoption, referring to firms' investment in and integration of AI tools into their operations, and passive AI exposure, referring to technological change at the industry level. For example, firm may face new competition, changes in demand, or shifts in production methods due to the AI impact elsewhere in its industry, even if a firm does not adopt AI itself. Distinguishing between these two channels is crucial for identifying the causal effects of AI and for drawing meaningful conclusions for policy.

Where data availability allows, the analysis also conducts cross-country comparisons, primarily between the United States and other countries with comparable disclosure requirements. This enables an assessment of whether institutional and regulatory differences shape how firms respond to AI.

The main research questions are: (i) how does AI adoption affect firm performance, labour demand, and corporate payout policy, and (ii) how do these effects differ across countries?

2 Research objectives and contribution

The first objective is to develop a new, longitudinal measure of AI adoption that clearly separates adoption from exposure. Existing studies measure AI adoption using employee resumes and job posting datasets (Babina, Fedyk, A. He, et al. 2024; Babina, Fedyk, A. X. He, et al. 2023) and cross-sectional adoption surveys (Aldasoro et al. 2026). In other cases, researchers look at the workforce AI exposure (Eisfeldt, Schubert, and Zhang 2023) and occupational tasks descriptions and patent filings (Webb 2020). I propose a firm-year measure of AI adoption based on firms' own disclosures, similar to the work of Basnet et al. (2025), but rather than an AI dictionary based approach, which count predefined keywords, I use LLMs to analyse firms disclosure to captures meaning and context, reducing the risk of classifying generic or frivolous mentions of AI words as adoption, allowing a direct and comparable assessment of adoption across firms.

The second objective is to provide new evidence on corporate financial responses to AI, with a focus on shareholder payout behavior. In particular, the dissertation examines whether the adoption of generative AI improve firm's performance, change labour demand and provide new investment opportunities that lead firms to reduce share buybacks. Prior research shows that AI adoption increases firm productivity and growth (Aldasoro et al.

2026; Babina, Fedyk, A. He, et al. 2024), reshape workforce composition toward a more educated and specialised workforce (Babina, Fedyk, A. X. He, et al. 2023), rising inequality (Acemoglu and Restrepo 2020) and affect firm values and risk (Eisfeldt, Schubert, and Zhang 2023; Basnet et al. 2025). However, the impact of AI on payout policy remains largely unexplored. Understanding whether AI investment crowds out payouts is important for debates on innovation incentives, capital allocation and shareholders taxation.

The third objective is to explore cross-country differences in AI adoption and its effects. Existing cross-country evidence suggests that AI adoption and its effects vary with institutional settings (Aldasoro et al. 2026; Calvino and Fontanelli 2023), but remains limited by non-comparable adoption measures and uneven data availability across countries.

The motivation for this research is both policy driven and academic. From a policy perspective, AI is increasingly viewed as a general purpose technology with far reaching implications for productivity growth, labour markets, and firms’ financial behavior. Governments and international organisations are actively debating AI regulation, innovation subsidies, and the taxation of shareholder payouts. Despite this interest, there is still limited causal evidence on how AI affects firm level outcomes, particularly outside the United States and in relation to financial policies such as share buybacks. From an academic perspective, the dissertation contributes to a rapidly growing literature by improving measurement, separating adoption from exposure, and identifying a new financial channel through which AI may affect firms.

3 Data and empirical strategy

The dissertation focuses mainly on revenues, labour demand and share buybacks as firms’ outcomes and tests the following hypothesis:

1. Firms with higher pre-2022 AI adoption experience higher post-2022 performance
2. AI adoption leads to a reduction in share buybacks after 2022.
3. Passive AI exposure affects labour demand differently from active adoption.

The empirical analysis uses a difference in differences framework to estimate the causal effects of AI adoption and exposure on firm level outcomes. The main firm level dataset is drawn from Compustat, accessed through WRDS, which provides detailed longitudinal information on firm revenues, employment, investment, balance sheet items, and corporate payout policies, making it well suited for studying firms’ real and financial responses over time.

To measure active AI adoption, the dissertation constructs a firm level AI adoption index using textual analysis of corporate disclosures. For US firms, the analysis uses Form 10-K filings obtained from the SEC EDGAR archive. These filings include the Item1 (Business) and Item7 (MD&A) sections in which firms describe their business, strategy, investment decisions, and use of technology.

LLMs are used to analyse disclosure text and construct a firm-year AI adoption score for use in panel regressions. Comparing results across two models helps address concerns about model-specific bias and additional validation is conducted using human evaluation on a subset of texts.

To measure passive AI exposure, the dissertation incorporates the dataset by NAICS industry level developed by Felten, Raj, and Seamans (2021). This dataset provides measures of AI exposure, based on how susceptible industry tasks are to AI related automation.

The empirical model compares firm outcomes before and after November 2022 using a difference-in-differences design with two way fixed effects. Firm fixed effects control for time invariant firm characteristics, while time fixed effects control for common shocks affecting all firms. The main estimating equation interacts the post 2022 indicator with measures of AI adoption and AI exposure. This allows the effects of active adoption and passive exposure to be identified separately. Identification relies on the assumption that, in the absence of the GenAI shock, firms with different levels of adoption and exposure would have followed similar trends. This assumption will be assessed using pre-treatment balances tables and event-study specifications.

4 Feasibility and challenges

A first challenge concerns processing large volumes of text. Form 10-K filings are lengthy text files, The analysis requires extracting the relevant narrative sections, specifically Item1 and Item7, using regular expressions, while excluding irrelevant references. Batch processing and caching will address the large processing volumes.

The most important challenge relates to the computational demands of using LLMs. Although I use open source LLMs, these require substantial memory and processing resources, making large scale output impossible in “normal” computers if run locally. Free hosted platforms typically impose strict limits on request volume or token usage, which are not suitable for processing large firm-year panels, unless purchase under a fee. However, the analysis only requires inference, not model training, which reduces complexity. These challenges could be addressed using university supported computing resources or available cloud infrastructure.

Finally, cross-country analysis depends on the availability of comparable disclosure data. The US will be the core sample, countries in Europe and elsewhere will be explored where sufficiently standardised disclosure requirements are systematically available, allowing the same methodology to be applied.

References

- Acemoglu, Daron and Pascual Restrepo (2020). “Robots and Jobs: Evidence from U.S. Labor Markets”. In: *Journal of Political Economy* 128.6, pp. 2188–2244. DOI: 10.1086/705716.
- Aldasoro, Iñaki et al. (2026). *AI adoption, productivity and employment: Evidence from European firms*. eng. EIB Working Papers 2026-02. Luxembourg. DOI: 10.2867/1772538. URL: <https://hdl.handle.net/10419/335876>.
- Babina, Tania, Anastassia Fedyk, Alex He, et al. (2024). “Artificial intelligence, firm growth, and product innovation”. In: *Journal of Financial Economics* 151, p. 103745. ISSN: 0304-405X. DOI: <https://doi.org/10.1016/j.jfineco.2023.103745>. URL: <https://www.sciencedirect.com/science/article/pii/S0304405X2300185X>.
- Babina, Tania, Anastassia Fedyk, Alex X He, et al. (June 2023). *Firm Investments in Artificial Intelligence Technologies and Changes in Workforce Composition*. Working Paper 31325. National Bureau of Economic Research. DOI: 10.3386/w31325. URL: <http://www.nber.org/papers/w31325>.
- Basnet, Anup et al. (2025). “Analyzing the market’s reaction to AI narratives in corporate filings”. In: *International Review of Financial Analysis* 105, p. 104378. ISSN: 1057-5219. DOI: <https://doi.org/10.1016/j.irfa.2025.104378>. URL: <https://www.sciencedirect.com/science/article/pii/S105752192500465X>.
- Calvino, Flavio and Luca Fontanelli (2023). “A portrait of AI adopters across countries: Firm characteristics, assets’ complementarities and productivity”. In: *OECD Science, Technology and Industry Working Papers* 2023.02.
- Eisfeldt, Andrea L, Gregor Schubert, and Miao Ben Zhang (May 2023). *Generative AI and Firm Values*. Working Paper 31222. National Bureau of Economic Research. DOI: 10.3386/w31222. URL: <http://www.nber.org/papers/w31222>.
- Felten, Edward, Manav Raj, and Robert Seamans (2021). “Occupational, Industry, and Geographic Exposure to Artificial Intelligence: A Novel Dataset and Its Potential Uses”. In: *Strategic Management Journal* 42.12, pp. 2195–2217. DOI: 10.1002/smj.3286. URL: <https://doi.org/10.1002/smj.3286>.
- Webb, Michael (2020). “The Impact of Artificial Intelligence on the Labor Market”. In: *SSRN Electronic Journal*. DOI: 10.2139/ssrn.3482150. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3482150.